Fast and Robust Active Cooling / Vacuum Reset in a Transmon-Storage-Readout cQED System

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We compare four cooling pictures for a storage-transmon-readout circuit-QED device: a pulsed sequential ladder, a continuous resonant bright-state protocol, a continuous detuned Raman-like protocol, and an auxiliary eliminated storage-damping benchmark. On the present device abstraction, the continuous Raman-like protocol gives the best overall speed-versus-robustness tradeoff, the pulsed ladder is the fastest practical protocol and the best first experimental target, and the resonant bright-state path is the raw speed leader only when transmon population is not heavily penalized. The data also show that strengthening the readout bath moves the Raman-like protocol closer to the idealized autonomous-cooling benchmark, supporting a readout-dominated reservoir-engineering strategy as the best long-term direction.

I. INTRODUCTION

Active reset in bosonic cQED hardware is only useful if it removes entropy quickly without demanding exceptional qubit coherence. In a storage-transmon-readout architecture, that means using the transmon as a nonlinear mediator while shifting as much irreversibility as possible into the lossy readout mode. The central design question is therefore not just which protocol is fastest in an ideal model, but which architecture keeps working once the transmon has finite T_1, finite T_2, dephasing, imperfect reset, and higher-level leakage.

II. COMPARED SCHEME CLASSES

The study compares three physical schemes plus one auxiliary benchmark:

1. a pulsed ladder $|g, 0_r, n_s\rangle \leftrightarrow |f, 0_r, n_s - 1\rangle \leftrightarrow |g, 1_r, n_s - 1\rangle \xrightarrow{\kappa_r} |g, 0_r, n_s - 1\rangle$,
2. a continuous resonant bright-state protocol with large real intermediate $|f\rangle$ occupation,
3. a continuous detuned Raman-like protocol with reduced real transmon occupation,
4. and an auxiliary eliminated benchmark with effective storage damping $\dot{\rho} \approx \Gamma_{\mathrm{cool}} \mathcal{D}[a_s]\rho$.

Only the first three are direct native device replays within the local simulation framework; the last is included as a clearly labeled benchmark because the eliminated storage-only jump operator is not available as a native primitive.

III. MAIN RESULTS

Figure 1 shows representative single-photon trajectories and Fig. 2 compresses the main speed, robustness, and leakage comparison. The selected operating points are:

TABLE I. Scheme comparison summary. The auxiliary benchmark is included only as an idealized reference.

Scheme	τ_e (ns)	Robustness	Final $\langle n_s \rangle$
Pulsed ladder	11.0	0.969	0.0041
Continuous bright-state	13.0	0.965	0.0017
Continuous Raman-like	243.5	0.982	0.0003
Effective benchmark	535.0	0.999	0.0008

- pulsed ladder: e-fold cooling time 11.0 ns, final storage occupation 0.0041, final transmon excitation 0.0090;
- continuous bright-state: e-fold cooling time 13.0 ns, final storage occupation 0.0017, final transmon excitation 0.0360;
- continuous Raman-like: e-fold cooling time 243.5 ns, final storage occupation 0.0003, final transmon excitation 0.0194;
- auxiliary benchmark: e-fold cooling time 535.0 ns, final storage occupation 0.0008.

These numbers make the tradeoff clear. The pulsed ladder is fastest because each driven stage is short and number-resolved, but it still leaves non-negligible transmon excitation after the chosen readout ringdown. The resonant bright-state protocol cools storage quickly, yet it keeps the transmon substantially occupied throughout the window, which makes it calibration- and coherence-sensitive. The Raman-like protocol is much slower in raw e-fold time, but it ends with the smallest storage occupation among the physical schemes and achieves the highest overall robustness score, which is why it wins the overall comparison.

IV. VALIDATION

Three validation layers were applied before ranking the schemes. First, the baseline single-photon trajectories were checked against the expected limiting behavior: the pulsed ladder should transfer population in short steps, the bright-state protocol should show stronger real transmon occupation, and the Raman-like path should cool more slowly while suppressing intermediate occupation. Second, representative convergence checks were run against time-step and truncation choices, with the saved convergence tables showing that the reported ranking does not change under the tested numerical variations. Third, robustness sweeps over transmon T_1, transmon T_2, calibration offsets, imperfect reset, and thermal loading were used as the deciding metric rather than ideal-model speed alone.

The validation results support the central design hypotheses. Schemes with larger real transmon occupation are indeed faster but more fragile, while the more virtual Raman-like channel remains the strongest physical option once decoherence and calibration error are folded into the score. The readout-linewidth sweep further confirms the expected matching argument: increasing the readout bath strength helps both continuous protocols and especially benefits the Raman-like path, which is the architecture most closely aligned with an effective irreversible storage damping channel.

V. DISCUSSION

The study resolves the central physics questions as follows.

1. **Which scheme is fastest?** The pulsed ladder is the fastest practical physical scheme, with an e-fold time of 11.0 ns. The bright-state protocol is close behind at 13.0 ns.

2. **Which scheme is most robust to transmon decoherence?** The continuous Raman-like protocol is most robust overall, because it ends with much lower storage occupation than the other physical schemes while keeping transmon excitation below the bright-state level.

3. **Is real or virtual transmon population better?** Real population wins on raw speed, but the more virtual Raman-like path wins on the combined robustness metric.

4. **How important is the readout as a bath?** Very important. The readout-linewidth sweep in Fig. 3 shows that both continuous schemes improve as the readout bath is strengthened, with the Raman-like protocol moving closer to the idealized benchmark as κ_r increases.

The higher-Fock and mixed-state results support a practical compromise. The pulsed ladder remains the best first lab target because it uses already-validated number-resolved pulses and does not require a redesigned dissipative environment. The longer-term direction is nevertheless to engineer a stronger readout-dominated

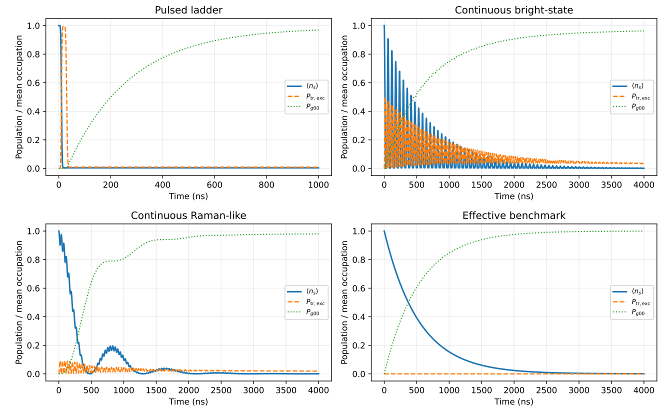

FIG. 1. Representative single-photon trajectories for the four compared schemes.

Raman-like protocol, because that is the path that most naturally approaches an effective irreversible storage jump operator.

VI. CONCLUSION

The best overall physical cooling architecture on the present device abstraction is the continuous Raman-like readout-assisted protocol, because it gives the best speed-versus-robustness balance once transmon decoherence and calibration sensitivity are included explicitly. The fastest practical scheme is still the pulsed ladder, which therefore remains the recommended first experimental target. The most important longer-term lesson is architectural rather than numerical: the readout resonator should be treated as the dominant engineered bath, while the transmon should be pushed toward a more virtual converter role whenever possible.

VII. LIMITATIONS AND FUTURE WORK

The sideband controls are still modeled as effective interactions rather than microscopic pump-derived Hamiltonians, and the continuous schemes were driven only with constant square overlaps rather than shaped counter-intuitive timing or open-system optimal control. Those limitations matter most for the long-term autonomous-cooling direction, not for the conclusion that the pulsed ladder is the safest first experiment on the current device abstraction.

Appendix A: Saved Artifacts

The machine-readable outputs are saved in `data/` and `artifacts/`, including candidate scans, robustness grids, convergence checks, and the final `study_results.json` summary.

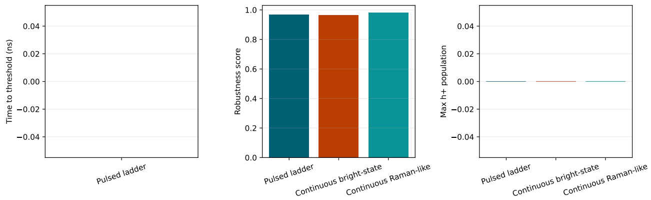

FIG. 2. Summary comparison across cooling speed, robustness score, and higher-level leakage.

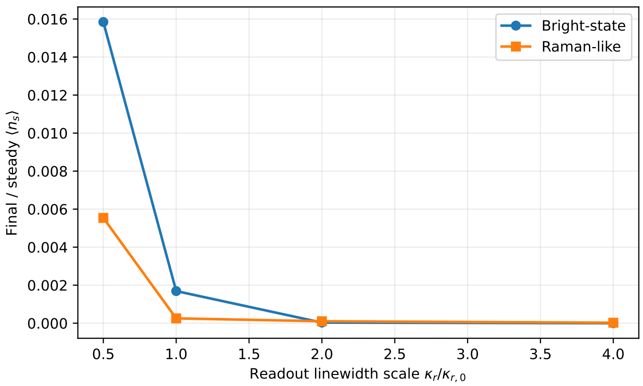

FIG. 3. Strengthening the readout bath improves both continuous schemes and especially benefits the Raman-like route, which is the clearest path toward the idealized autonomous benchmark.